

Research papers

On the incidence of meteorological and hydrological processors: Effect of resolution, sharpness and reliability of hydrological ensemble forecasts

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ABSTRACT

Meteorological and hydrological ensemble prediction systems are imperfect. Their outputs could often be improved through the use of a statistical processor, opening up the question of the necessity of using both processors (meteorological and hydrological), only one of them, or none. This experiment compares the predictive distributions from four hydrological ensemble prediction systems (H-EPS) utilising the Ensemble Kalman filter (EnKF) probabilistic sequential data assimilation scheme. They differ in the inclusion or not of the Distribution Based Scaling (DBS) method for post-processing meteorological forecasts and the ensemble Bayesian Model Averaging (ensemble BMA) method for hydrological forecast post-processing. The experiment is implemented on three large watersheds and relies on the combination of two meteorological reforecast products: the 4-member Canadian reforecasts from the Canadian Centre for Meteorological and Environmental Prediction (CCMEP) and the 10-member American reforecasts from the National Oceanic and Atmospheric Administration (NOAA), leading to 14 members at each time step. Results show that all four tested H-EPS lead to resolution and sharpness values that are quite similar, with an advantage to DBS + EnKF. The ensemble BMA is unable to compensate for any bias left in the precipitation ensemble forecasts. On the other hand, it succeeds in calibrating ensemble members that are otherwise under-dispersed. If reliability is preferred over resolution and sharpness, DBS + EnKF + ensemble BMA performs best, making use of both processors in the H-EPS system. Conversely, for enhanced resolution and sharpness, DBS is the preferred method.

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1. Introduction

Ensemble forecasting is a probabilistic approach that favours the use of multiple model runs with different inputs, initial conditions and/or model physics. Widely used ensemble meteorological forecasts, such as those from the CCMEP (Pellerin et al., 2003), the National Centers for Environmental Prediction (NCEP) (Toth and Kalnay, 1993), or the European Centre for Medium-Range Weather Forecasts (ECMWF) (Buizza et al., 2007), are known to often be under-dispersive and locally biased (Schaake et al., 2007). Biases occur not only in relation to the mean, but also in higher statistical moments of the predictive distribution, limiting the capacity of the ensemble to reliably depict the true uncertainty of the forecast (Eckel and Walters, 1998). As a consequence, usage of raw forecast

values may not exploit the full potential of information on predictive uncertainty inherent to the models (Schaake et al., 2007). Retrieving that information requires a meteorological processor to address model biases and dispersion errors of the ensemble members.

For similar reasons, in addition to the post-processing of meteorological data, post-processing of hydrological ensemble forecasts may also be required to guarantee the issued ensemble members are unbiased and exhibit the appropriate spread (Madadgar et al., 2014). Very few studies have previously addressed the comparison between meteorological and hydrological processors (e.g. Zalachori et al., 2012; Roulin and Vannitsem, 2015).

Many popular methods for post-processing ensemble meteorological forecasts are described by Wilks (2006a). Amongst the methods compared are: ensemble Bayesian Model Averaging (ensemble BMA) (Raftery et al., 2005; Baran, 2010), Non-homogeneous Gaussian regression (NGR) (Gneiting et al., 2005), ensemble dressing (Roulston and Smith, 2003; Wang and Bishop, 2005), and logistic regression (Wilks, 2006b). Wilks concluded that

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the most promising methods were NGR, logistic regression, and ensemble dressing. However, ensemble BMA performed well for post-processing of multimodel ensembles. Yang et al. (2017) compared the BMA and heteroscedastic censored logistic regression (HCLR) methods for post-processing of the 11-member Global Ensemble Forecast System reforecasts, where the HCLR method slightly outperforms the BMA. Recently, Khajehei and Moradkhani (2017) used a Bayesian ensemble post-processing approach based on copula functions to improve the reliability of ensemble meteorological forecasts and this method provided a reliable and unbiased ensemble forecast. There few studies which have addressed the post-processing of ensemble hydrological forecasts (Roulin and Vannitsem, 2015; Boucher et al., 2015).

The objective of this study is to obtain hydrological ensemble forecasts with best possible performance and resolution, without reducing the reliability, through the use of both meteorological and hydrological statistical post-processors. Meteorological post-processing is performed by applying the Distribution Based Scaling method (DBS: Rana et al., 2014; Wetterhall et al., 2012), which focuses on reducing the bias of ensemble meteorological reforecasts – it is assumed here that there is no need to improve on the under or over dispersion problems, since that task is accounted by the hydrological processor. Hydrological post-processing is implemented using the ensemble BMA. Raftery et al. (2005) proposed this method to adjust forecast ensemble and generate predictive probability density functions (PDFs) for future weather quantities (Berrocal et al., 2007). The ensemble BMA predictive PDF is defined as a weighted average of predictive PDFs associated with each individual ensemble member, with weights consistent with the member's relative skill.

Two meteorological reforecast products are combined in order to operate with different meteorological ensemble prediction systems: the 4-member Canadian reforecasts from CCMEP and the 10-member American reforecasts from NOAA, leading to 14 members at each time step. Working with reforecast data allows exploration of a long time span and diverse climatic conditions.

The Ensemble Kalman Filter (EnKF), a probabilistic sequential data assimilation technique, is also applied to improve the reliability, sharpness and resolution of the ensemble hydrological reforecasts (e.g. Abaza et al., 2014a; Abaza et al., 2014b; Abaza et al., 2017), prior to hydrological post-processing. Other studies (e.g. Parrish et al., 2012; DeChant and Moradkhani, 2014; Madadgar and Moradkhani, 2014) combine ensemble data assimilation and post-processing methods in order to improve the reliability and performance and reduce the uncertainty of hydrological forecasting. They were shown to outperform applications based on a single component (data assimilation or post-processing).

This paper is organized as follows. In Section 2, we describe the data and watersheds dataset considered herein. In Section 3, the implementation procedure is explained. In Section 4, results and discussion are exposed. Finally, conclusions are presented in Section 5.

2. Material and methodology

2.1. Study area and data

Three watersheds located in the province of Québec (Canada) have been considered in this study (Fig. 1). They all drain a large area, ranging from 9 426 km² for the Gouin watershed to 17,109 km² for the Outardes-4 watershed. Strategic management decisions are based on hydrological forecasts produced for these sites, namely to prevent flooding damages and avoid operating losses. Therefore, producing skillful hydrological forecasts for these watersheds is of primary importance.

Two meteorological reforecast products are combined: the Global Ensemble Prediction System (GEPS) reforecasts, which are operationally issued by the CCMEP, based on the Global Environmental Multiscale (GEM) model (Girard et al., 2014) And the GEPS reforecasts, which consist of four 50-km members issued once a week over a 32-day horizon, for years spanning from 1995 to 2012 – this span has been extended to 2014 after the end of the calculations presented herein. The four members of the GEPS product are selected from the 20 members of the GEM simulations and member selection varies from year to year.

According to Charron et al. (2010), the deep convection parameterization in GEM is considered has an important impact on the skill and bias of the precipitation forecast. Two parameterizations of comparable skill but differing bias are used in the operational ensemble forecasting system, and each of them is used in 10 members of the 20 member ensemble: odd-numbered members use the Kuo parameterization and even-numbered members use the Kain and Fritsch parameterization (see http://collaboration.cmc.ec.gc.ca/cmc/ensemble/doc/info_geps_e.pdf). Unfortunately, computing resources are insufficient to run a 20-member ensemble reforecast; only four members are available in the reforecast dataset. To be as coherent as possible with its regular 20-member forecast product, four reforecast members are alternately chosen from the initial 20-member GEM runs, and in each case two odd-numbered and two even-numbered members are chosen. Hence, each member includes the same number of times over the years. For example, 1995 exploits members 1, 6, 11, and 16, while 1996, members 2, 7, 12, and 17. Overall, 72 reforecasts are thus issued each time: 4 members over 18 years.

The second product consists of the second generation of NOAA's Global Ensemble Forecast System (GEFS-v2): 11 members (10 + 1 control) over a 16-day horizon, issued each day from December 1984 until present (Hamill et al., 2013). The horizontal resolution of GEFS-v2 is 50 km up to day 8, and subsequently 70 km.

The daily climate input to the hydrological model (precipitation and temperature) also originates from NOAA. It has a grid resolution of about 6 km, covering most of Canada, Mexico, and conterminous U.S. (CONUS), and extends from January 1950 to December 2013 (Livneh et al., 2015). All the reforecast data defined above have been interpolated to each basin, producing an average amount for the watershed as required by the lumped model used in this study (GR4J). In the case of precipitation and temperature observation data, an average of the grid points which are inside of the basin has been used as an average of the input data.

The daily hydrological data (streamflow), provided by Hydro-Québec, extends from 1995 to 2012. It has been split evenly for the calibration and validation of the hydrological model.

2.2. Meteorological processor

A meteorological processor is often required to obtain a maximized resolution, subject to a reliable description of the forecast uncertainty (Gneiting and Raftery, 2007). A range of tools exist for this (e.g. Wilks, 2006a), such as the ensemble BMA, Ensemble Model Output Statistics (EMOS), kernel dressing, and parametric statistical methods (e.g. Scheuerer and Hamill, 2015). Considering that a hydrological processor is also used in this study, a meteorological processor was selected that would focus exclusively on the correction of the local bias: the Distribution Based Scaling (DBS) method (Yang et al., 2010). The fact that DBS does not improve the dispersion of the ensemble is not detrimental to the study since a hydrological processor is available for this purpose at the end modelling chain, if needed. The DBS approach operates directly on the precipitation outputs in order to preserve the statistical distribution of the observed precipitation for the reference period (Seaby et al., 2013). Application of the DBS consists of two steps:



Fig. 1. The location of the three watersheds highlighted in dark grey (source: Hydro-Québec). The inset shows the 90 watersheds managed by Hydro-Québec.

- (1) Calculate a threshold from the reforecast and observed data to define a rainy days minimum quantity. Both reforecast and observed precipitation data that have values below this threshold are then removed, and the data above this threshold, are preserved. According to Haylock et al. (2006), DBS applies a gamma distribution because of its ability to represent the typically asymmetrical and positively skewed distribution of daily rainfall intensities. The PDF of the 2-parameter gamma distribution considered herein is given by:

$$f(y|\alpha, \beta) = \frac{y^{\alpha-1}}{\beta^{\alpha}\Gamma(\alpha)} \exp\left(-\frac{y}{\beta}\right), y, \alpha, \beta > 0 \quad (1)$$

where α is the shape parameter, β is the scale parameter, y is the daily precipitation (mm/day), and $\Gamma(\alpha)$ is the gamma function. The maximum likelihood estimation (MLE) method is used to estimate the shape and scale parameters of non-zero precipitation events (Wilks, 2006b). Small precipitation amounts were left unchanged.

- (2) Estimate the DBS parameters for each watershed and the CDFs of the reference and observed periods, in order to determine the adjusted daily precipitation value as follows:

$$Y_{DBS} = F^{-1}[F(Y_{ref}|\alpha_{ref}, \beta_{ref})|\alpha_{obs}, \beta_{obs}] \quad (2)$$

where Y_{DBS} and Y_{ref} are the bias corrected and reference reforecast precipitation at selected time and location in space, respectively. F is the CDF of the gamma distribution, and obs and ref refer to parameters estimated based on the observed and reference periods, respectively. Finally, F^{-1} is the inverse CDF of the gamma distribution.

2.3. Hydrological processor

Most hydrological ensemble prediction systems (H-EPS) fail to account for all sources of uncertainty (Thibault et al., 2016), leading to under-dispersed forecasts. A hydrological processor is thus deemed necessary in most situations, not only to reduce the bias, but more importantly to correct for under-dispersion. The ensemble Bayesian Model Averaging (ensemble BMA) (Raftery et al., 2005; Wilson et al., 2007; Sloughter et al., 2007) was built to improve the reliability and resolution of multimodel ensemble hydrological reforecasts. The objective of this method is to calibrate and provide sharp predictive ensemble PDFs (Sloughter et al., 2007).

The ensemble BMA post-processing method of Raftery et al. (2005) has been used in this study to adjust the ensemble streamflow reforecasts. If we consider $r_1, \dots, r_N$ to be the ensemble streamflow reforecasts, for a streamflow quantity z , at a given time and location, the conditional predictive PDF associated to one ensemble member is presented by a gamma distribution $f_k(z|r_k)$. The ensemble BMA model for the predictive PDF of the streamflow quantity z can be expressed as a finite mixture model, given as follows:

$$p(z|r_1, \dots, r_N) = \sum_{k=1}^N w_k f_k(z|r_k) \quad (3)$$

with ensemble BMA weights $w_1, \dots, w_N$ that are nonnegative and sum to 1, r_k is the original forecast from the member k , and N is the number of ensemble members. The ensemble BMA weights may provide a crude assessment of the relative resolution of the ensemble member models during the training period.

The mean and the variance of the gamma distribution, as a function of the original reforecast r_k , are estimated as follows:

$$\mu_k = r_k$$

$$\sigma_k^2 = br_k + c_0$$

where b , c_0 , μ_k and σ_k depend upon the parameters of the gamma distribution (Eq. (1)):

$$\mu_k = \alpha_k \beta_k$$

$$\sigma_k^2 = \alpha_k \beta_k^2$$

In this study, the R package ensemble BMA (Fraley et al., 2011) is applied to estimate all parameters.

For a given training sample and based on the gamma distribution, this package uses the maximum likelihood approach (ML) to estimate the weights and the parameters (b , c_0) of the ensemble BMA model. The Expectation-Maximization (EM) iterative algorithm (Dempster et al., 1977) is employed to find point-estimates for all parameters

2.4. Hydrological model

A daily lumped conceptual rainfall-runoff model (GR4J; Perrin et al., 2003) is utilised in this study, as it was deemed comparable to the semi-distributed operational model in used in the province of Québec (Thibault and Anctil, 2015). GR4J is structured around four parameters and two stores: production and routing. The levels of the two stores are considered as state variables.

The CemaNeige snow module developed by Valéry et al. (2014) simulates snow accumulation and melting. The latter is added to rainfall to form the precipitation input to the hydrological model. More information about GR4J and CemaNeige are given by Abaza et al. (2014a).

A calibration process identical to Abaza et al. (2014b) is performed. Performance of calibration and validation periods is presented in Table 1 which reveals that the model behaves somehow better in calibration than in validation period. In the case of the Gouin watershed, GR4J fails to preserve the good performance obtained in the calibration, in good part due to the difference of climatic conditions between both periods.

Figs. 2 and 3 illustrate the climatic and hydrologic conditions of the three selected watersheds, in calibration and validation, respectively. The GR4J model succeeds to reproduce rainfall-driven peaks for most of the calibration and validation periods on Baskatong and Outardes 4 watersheds, but to a lesser extent on the Gouin watershed.

2.5. Ensemble Kalman Filter

The Ensemble Kalman Filter (EnKF; Evensen, 2003; Evensen, 2009) is a sequential data assimilation technique that is considered suitable for hydrological applications. It has the ability to update

states (or more rarely the parameters) of a model when new observation becomes available (Xie and Zhang, 2010). In EnKF, the set of state vectors propagates with time through the hydrological model and each state vector is provided for each ensemble member simulation, expressed as:

$$x_{t+1}^i = f(x_t^i, u_t, w_t), \quad i = 1, \dots, N \quad (4)$$

where x_{t+1}^i is the forecasting state of the member i at time $t + 1$, x_t^i is the forecasting state of the member i at time t , w_t is the model uncertainties and u_t is the vector of model inputs like precipitation and temperature. In EnKF, the inputs and outputs need to be perturbed through the addition of noise in order to represent uncertainty (Burgers et al., 1998), creating an ensemble of observations for which the mean equals to the original observation over a given time with predetermined covariance. The new forecasted states x_{t+1}^i are adjusted as a function of Kalman gain K_{t+1} as expressed by:

$$x_{t+1}^i = x_{t+1}^i + K_{t+1}(z_{t+1}^i - H(x_{t+1}^i)) \quad (5)$$

where $H(x_{t+1}^i)$ is the simulated forecast at time $t + 1$, and z_{t+1}^i is the i th value of perturbed observation (streamflow). If the observations are a nonlinear combination of state variables, the Kalman gain is computed as:

$$K_{t+1} = \sum_{i=1}^{xz} \left[\sum_{t+1}^{zz} + \sum_{t+1}^z \right]^{-1} \quad (6)$$

where $\sum_{t+1}^{xz}$ is considered as the cross covariance of the state variables x_{t+1}^i and prediction $H(x_{t+1}^i)$, $\sum_{t+1}^{zz}$ is the forecast error covariance matrix of the forecast $H(x_{t+1}^i)$, and $\sum_{t+1}^z$ is the covariance of the observations.

2.6. Verification

It is important to evaluate the quality of meteorological and hydrological ensemble prediction systems in order to quantify their effectiveness (Perreault et al., 2017). Scores used for forecast verification are diagnostic tools designed to assess resolution, sharpness, skill, and/or reliability.

The goal of the probabilistic forecasting is to “maximize the sharpness of the predictive distributions subject to calibration” (Gneiting et al., 2007). Here, calibration (reliability) refers to the statistical consistency between the predictive distribution and the observation. For instance, a forecast is reliable if it is unbiased and not under-dispersed or over-dispersed. Resolution is defined as the ability of the forecast model to correctly predict the occurrence or non-occurrence of events (Demirel et al., 2013). Sharpness refers to the concentration of the predictive distribution and it is a property of the forecasts only. Finally, skill informs on the accuracy relative to a reference forecast methodology, usually the local climatology.

In this study, we primarily consider some scores used to evaluate the quality of probabilistic forecasts, such as the Continuous

Table 1
Calibration and validation performance values: T = number of time steps; t = time step index; Q_{sim} = simulated streamflow; Q_{obs} = observed one; and $\overline{Q_{obs}}$ = mean of the observed flows.

	Calibration period		Validation period	
	Nash-Sutcliffe (NSE)	Root Mean Square Error (RMSE) (mm)	Nash-Sutcliffe (NSE)	Root Mean Square Error (RMSE) (mm)
	$NSE = 1 - \frac{\sum_{t=1}^T (Q_{obs,t} - Q_{sim,t})^2}{\sum_{t=1}^T (Q_{obs,t} - \overline{Q_{obs}})^2}$	$RMSE = \sqrt{\frac{\sum_{t=1}^T (Q_{obs,t} - Q_{sim,t})^2}{T}}$		
Baskatong	0.84	0.7	0.83	0.7
Outardes	0.83	0.58	0.77	0.74
4				
Gouin	0.71	0.79	0.55	1.04

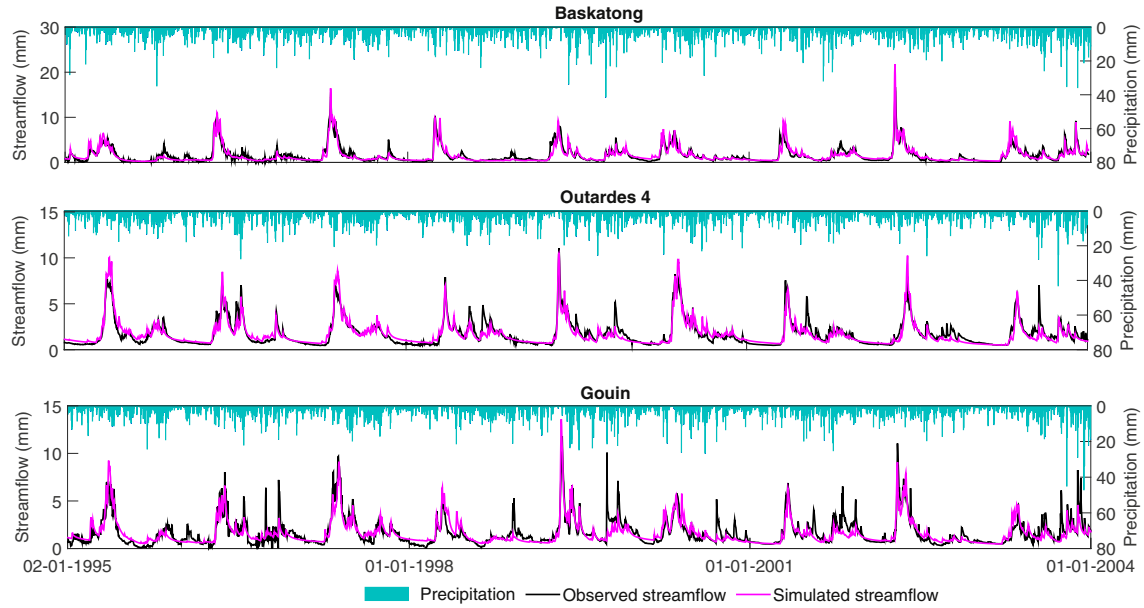


Fig. 2. Climatic and hydrologic conditions of all the three watersheds in the calibration period.

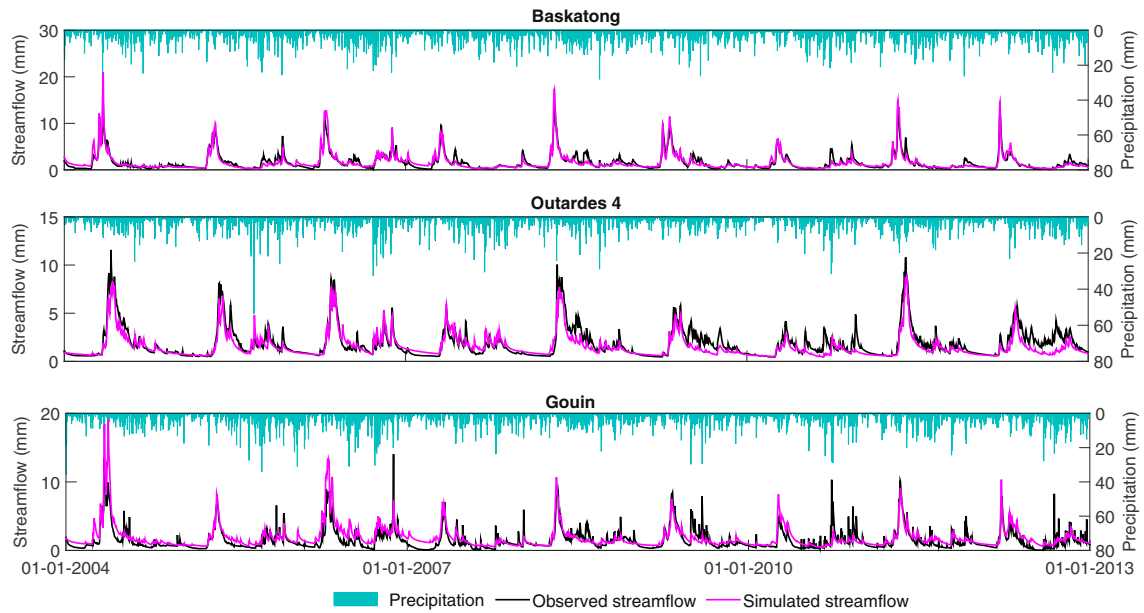


Fig. 3. Climatic and hydrologic conditions of the three watersheds in the validation period.

Ranked Probability Score (CRPS) proposed by Matheson and Winkler (1976). Such a scoring rule addresses calibration and resolution simultaneously, and possesses the same units as the observed quantity of interest. Therefore, it is an interesting summary measure of predictive performance. Assuming F_t , the predictive distribution function at time t , and y_t , the corresponding observation, the CRPS is defined as:

$$CRPS(F_t, y_t) = \int [F_t(x) - \mathbf{I}_{[y_t, +\infty)}(x)]^2 dx \quad (7)$$

where $\mathbf{I}_A(x)$ is the well-known indicator function defined as:

$$\mathbf{I}_A(x) = \begin{cases} 0 & \text{if } x \notin A \\ 1 & \text{if } x \in A \end{cases}$$

An ensemble is considered to be perfect forecasts when the CRPS approaches zero. The CRPS score can be decomposed into reliability component and CRPS “potential” component (Hersbach, 2000). According to the latter, the reliability term assesses how much the forecasts are reliable. The CRPS “potential” ($CRPS_{pot}$) term defines the lowest value of CRPS that can be obtained if the ensemble were perfectly reliable (Hersbach, 2000). The lower this value the more the ensemble has a high resolution (good forecast). The reliability and $CRPS_{pot}$ components have been evaluated separately in this study in addition to the total CRPS. Following Hersbach (2000), the decomposition may be written as follows:

$$CRPS = \text{Reliability} + CRPS_{pot} \quad (8)$$

where

$$\text{Reliability} = \sum_{i=0}^N g_i (o_i - p_i)^2 \quad (9)$$

$$\text{CRPS}_{\text{pot}} = \sum_{i=0}^N g_i o_i (1 - o_i) \quad (10)$$

with g_i as the average width of the bin i (Euclidean distance between consecutive ensemble members x_i and x_{i+1} for $0 < i < N$, and Euclidean distance between the observation and the outliers for $i = 0$ and $i = N$), o_i is the average frequency that the observation is less than the middle of bin i , N is the ensemble size, and p_i is the fraction i/N . The objective is to obtain an ensemble forecast with low values for both components of the CRPS (Reliability and CRPS_{pot}).

The ratio δ and reliability diagram assess reliability. The ratio δ computes the normalized deviation of the rank histogram from flatness. It informs on the dispersion of the members of the probabilistic forecasts (Hamill and Colucci, 1997). The rank of the observation with respect to the ensemble is first evaluated. In the rare case for which the observation perfectly matches one or more members of the ensemble, a random selection is used to discriminate between possible ranks. The ratio δ is expressed as follows (Candille and Talagrand, 2005):

$$\delta = \frac{\Delta}{\Delta_0} \quad (11)$$

where Δ corresponds to the deviation of the histogram from flatness:

$$\Delta = \sum_{k=1}^{N+1} \left(s_k - \frac{M}{N+1} \right)^2 \quad (12)$$

where s_k is the number of elements in each interval k of the rank histogram ($k \in [1, N+1]$), N is the number of members of the ensemble, and M is the number of forecasts in the archive. $\frac{M}{N+1}$ is the expectation of s_k in the case of a reliable system. Δ_0 is the ratio that can be obtained by a perfect reliable system:

$$\Delta_0 = \frac{MN}{N+1} \quad (13)$$

A ratio δ value greater than 1 indicates unreliability, more so as the value increases.

The reliability diagram (Wilks, 1995) compares the observed relative frequency for some classes of distribution with the theoretically expected value. This means that the observation should be contained on average 30% of the time in 30% of the central point of the distribution, 50% of the time in 50%, and so on. The reliability diagram tracks this frequency according to the theoretical probability. Hence, for each confidence interval (10–90%), the observed relative frequency is computed, to have an idea if the observation is within the appropriate range. This procedure should be repeated for all forecasts in the archive (reforecasts in this study). A reliable EPS should, in theory, lead to the representation of a line of slope 1 and intercept 0. A lower curve indicates under-dispersion and a higher one, over-dispersion.

Another simple and effective score, called the average width of the prediction intervals (WPI) (Qu et al., 2017), is used to assess the sharpness of ensemble hydrological reforecasts. This score represents the average width of a 90% prediction interval and is defined as follows:

$$\text{WPI} = \frac{1}{M} \sum_{m=1}^M (f_m^u - f_m^l) \quad (14)$$

where f_m^u and f_m^l are the upper and lower bound of the 90% prediction interval, respectively. The smaller the value of WPI, the better the forecast sharpness.

The skill of the ensemble hydrological reforecast is evaluated using the Continuous Ranked Probability Skill Score (CRPSS) which compares the average CRPS (MCRPS) of the verified forecast to a reference unskilled prediction. Climatology is used for the reference forecast. It is computed as follows:

$$\text{CRPSS} = 1 - \frac{\text{MCRPS}_{\text{forecast}}}{\text{MCRPS}_{\text{climatology}}} \quad (15)$$

where the average CRPS of the reference and the verified forecasts are respectively denoted by the $\text{MCRPS}_{\text{climatology}}$ and $\text{MCRPS}_{\text{forecast}}$. An ensemble is perfectly skillful when the CRPSS is equal to 1.

3. Experimental setup

The DBS is used to first process the GEPS/GEFS-v2 14-member precipitation reforecast data. This method has been used to ensure precipitation inputs are compatible with our hydrological model with the goal of improving the hydrological forecasts, being conscious that DBS reduces variability. It is hypothesized that reducing the local bias of the precipitation could improve the hydrological reforecasts that are issued using the GR4J model in conjunction with EnKF streamflow assimilation, which is expected to improve the reliability of the reforecasts by accounting for the initial conditions uncertainty. Note that in order to remove as much as possible the influence of the air temperature on this study, observed temperatures are used to replace the reforecasted ones. Note also that since CCMEP issues GEPS reforecasts once a week, only one out of seven GEFS reforecasts is exploited. Finally, ensemble BMA is applied to the raw hydrological forecasts. Ensemble BMA has the ability to improve the reliability with a risk of reducing the resolution and sharpness.

The efficiency of the DBS, as for any other statistical approaches, is dependent on the size of the training sample available to estimate the parameters. The training dataset is constructed using a moving window extending over 5 reforecasts (14 members times 18 years times 5 reforecasts) centred around the date of interest. In order to reduce the risk of overfitting, members coinciding with the date of interest and with the two following weeks are excluded from the training dataset, which at the end consists of a moving window of 14 times 87 reforecasts (14 members' times 18 years times 5 reforecasts minus 3 reforecasts).

The available Canadian database extends over 52 weeks, from January 7 to December 31, 1995 to 2012; but, because the moving window extends over 5 weeks, in total 864 reforecasted events are post-processed: 48 weekly reforecasts times 18 years.

Preparation and organization of the hydrological post-processing are similar to that of the meteorological one (DBS). The primary difference is that, for the ensemble BMA, 44 weekly hydrological reforecasts are produced over 18 years, since we have again to ignore the first two, as well as the last two, weeks of the 48 weekly hydrological reforecasts.

The EnKF implementation is similar to that of Abaza et al. (2014a): 50 EnKF members and individual hyper-parameters for the perturbation of the inputs (precipitation and temperature), output (streamflow), and the two states (production and routing stores levels) of the GR4J model.

The principal methodological elements are summarized by Fig. 4. It includes the DBS meteorological processor for the ensemble meteorological reforecasts, the EnKF assimilation of the observed streamflow, and the ensemble BMA hydrological processor of the ensemble hydrological reforecasts.

In order to thoroughly assess the influence of meteorological and hydrological processing, four scenarios are assessed:

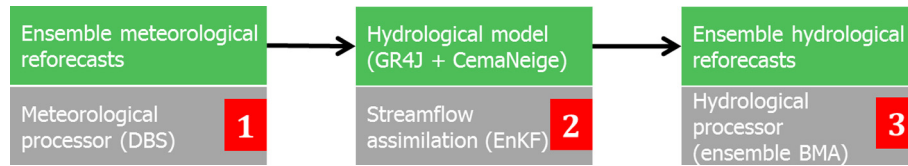


Fig. 4. The different components of the methodology adopted in this study.

- (1) *EnKF*: data assimilation alone;
- (2) *DBS + EnKF*: meteorological processing and data assimilation;
- (3) *EnKF + ensemble BMA*: data assimilation and hydrological processing;
- (4) *DBS + EnKF + ensemble BMA*: meteorological processing, data assimilation, and hydrological processing.

The number of members varies from one scenario to another. The first two scenarios hold 700 members (50 *EnKF* members times 14 raw or *DBS* members). The last two scenarios consist of 50 members which are randomly generated by the ensemble BMA, based on the weights, shape, and scale parameters of the members used as inputs of the latter. Since it is very computationally demanding to apply this methodology to 700-member ensembles, we restricted the ensemble to 50 members by randomly choosing one meteorological member per *EnKF* member, following Abaza et al. (2014b). The 50 members used as inputs to the ensemble BMA model are assumed to be exchangeable; however, they have the same weight and parameter values.

4. Results

4.1. Meteorological assessment

Fig. 5 illustrates both components of the MCRPS and the MCRPS of the 14-member raw and processed precipitation reforecasts. *DBS* provides a substantial improvement in terms of the $MCRPS_{pot}$ component of the MCRPS over all three watersheds, especially for lead times of 1–6 days. This tends to indicate that the raw product

is locally biased and that the *DBS* processor improves the resolution and achieves the desired outcome. In terms of reliability, Fig. 5 shows some improvements for up to about day 4, followed by deterioration for longer horizons. Such results are expected from the *DBS* since it is not designed to improve problems of under or over dispersion. This is confirmed in Fig. 6, some deterioration of the ratio δ is noticeable for up to day 6, followed by an improvement for longer horizons.

On the contrary, reducing the bias of the individual members has the general consequence of diminishing the ensemble spread. The MCRPS results are dominated by its $MCRPS_{pot}$ component, which is quite large compared to its reliability counterpart, especially for longer horizons.

The role of the meteorological ensemble forecasts in a hydrological ensemble prediction system is to provide information on the precipitation uncertainty (Schaake et al., 2007). Of course, one would like the meteorological uncertainty description to be as reliable as possible, but, as already mentioned, it is hypothesized here that subsequent tools within the hydrological ensemble prediction system may compensate for some (if not most) of the errors in that respect.

Table 2 provides the mean values of the *DBS* parameters for Day 3, per season, watershed, and meteorological prediction system. One can see that the parameters do not vary considerably from one season to the other, not from one watershed to the other, opening up the possibility of application to neighbouring ungauged watersheds. This issue, of course, needs further investigation since only three watersheds are available for the current study. It is also noteworthy that the forecast error varies from one season to the other and is largest in summer (not shown here), probably due

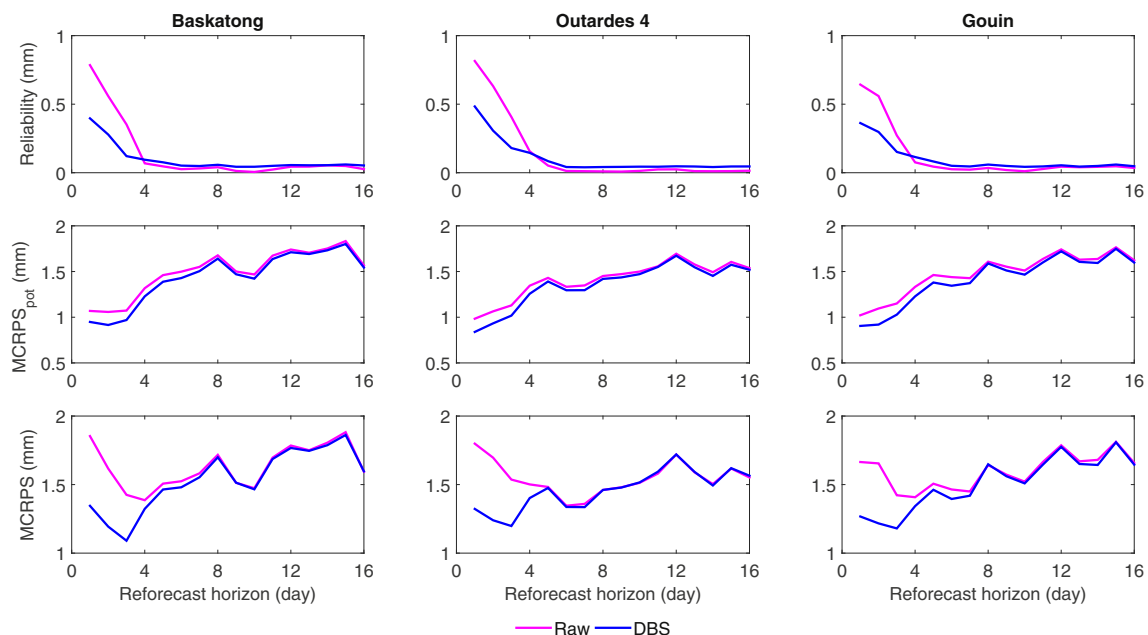


Fig. 5. Values of the MCRPS and of its two components for the raw and *DBS*-corrected precipitation rates, as a function of the reforecast horizon.

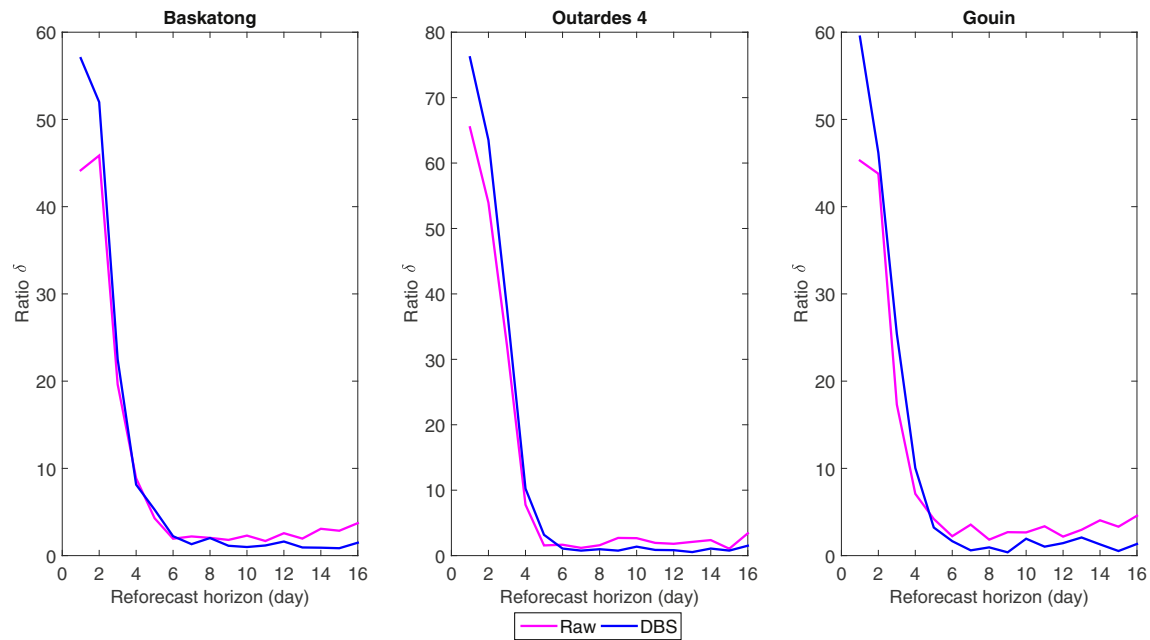


Fig. 6. Ratio δ values of the raw and DBS corrected precipitation rates, as a function of the reforecast horizon.

Table 2

Day 3 mean value of DBS gamma distribution parameters across the four seasons, three watersheds, and two meteorological prediction systems.

		Basketong		Outardes 4		Gouin	
		α	β	α	β	α	β
Winter	Observed	0.65	3.50	0.61	3.18	0.70	2.98
	4-member GEPS	0.49	5.27	0.49	3.30	0.54	4.00
	10-member GEFS v2	0.53	6.87	0.44	6	0.51	6.24
Spring	Observed	0.43	5.85	0.48	4.57	0.48	5.00
	4-member GEPS	0.42	8.33	0.51	5.54	0.41	8.33
	10-member GEFS v2	0.46	11.5	0.48	8.82	0.44	12.62
Summer	Observed	0.58	5.53	0.70	4.83	0.70	5.05
	4-member GEPS	0.40	10.82	0.67	7.09	0.44	10.3
	10-member GEFS v2	0.40	12.98	0.46	12.88	0.33	17.93
Autumn	Observed	0.61	4.23	0.62	4.85	0.69	4.20
	4-member GEPS	0.41	8.23	0.52	7.12	0.49	6.76
	10-member GEFS v2	0.45	9.65	0.43	11.9	0.38	11.7

to convective events that are not fully accounted for by the meteorological models. DBS is not able to compensate for such limitation to the forecasts. Fig. 7 shows the Empirical Cumulative Distribution Functions (CDF) of the observation and ensemble mean of raw and DBS corrected precipitation rates for the 2-day reforecast horizon. We can deduct from this Figure that the DBS succeeded in reducing the bias and preserving the statistical distribution of the observed precipitation. The same results were obtained for the other reforecast horizons (not shown).

4.2. Hydrological assessment

Four hydrological scenarios are investigated, as described in Section 3. They differ by the use or not of a meteorological processor (DBS) and of a hydrological processor (ensemble BMA), but all share the same hydrological model (GR4J) and data assimilation tool (EnKF): (1) EnKF; (2) DBS + EnKF; (3) EnKF + ensemble BMA; and (4) DBS + EnKF + ensemble BMA. 864 hydrological reforecasts are constructed for the first two scenarios, based on 48 precipitation reforecasts over 18 years, while 792 hydrological reforecasts are calculated for the other two scenarios, 44 precipitation refore-

casts over 18 years. To be coherent in the following comparison, the same 792 hydrological reforecasts are retained to assess all scenarios.

The MCRPS and its components are drawn in Fig. 8 for the EnKF (reference) scenario. One can see that the MCRPS increases with the reforecast horizon and is dominated by its $MCRPS_{pot}$ component, as its reliability component diminishes with the reforecast horizon. This H-EPS accounts for meteorological uncertainty and initial conditions uncertainty, but does not directly account for structural (hydrological model) uncertainty. Some processing is thus needed to compensate for the latter.

Performance gains (losses) of the three other scenarios, when compared to the EnKF one, are presented in Fig. 9. Usage of a meteorological or hydrological processor generally improves the hydrological forecasts with respect to the MCRPS. Fig. 9 reveals that DBS (green) substantially improves the $MCRPS_{pot}$ or more specifically the hydrological resolution over the EnKF scenario, especially up to day 6, while its contribution to reliability is not as systematic, as expected, since it only aims at correcting bias. Fig. 9 also reveals that ensemble BMA (red and blue) greatly improves the hydrological reliability, but at the expense of resolution, particularly for the

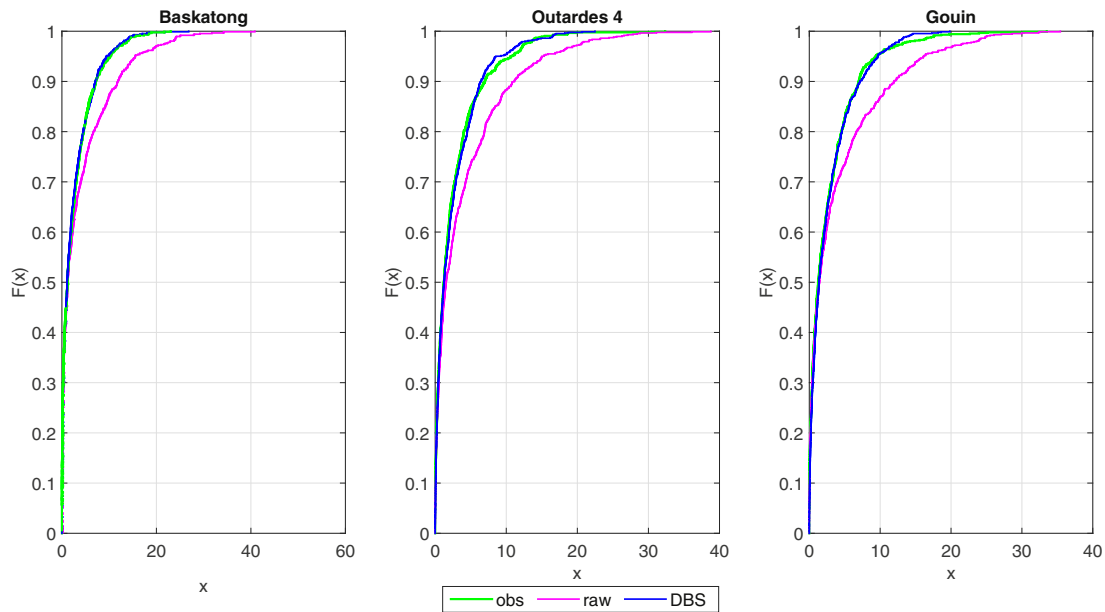


Fig. 7. Empirical Cumulative Distribution Functions (CDF) of the observation and ensemble mean of raw and DBS corrected precipitation rates for 2-day reforecast horizon.

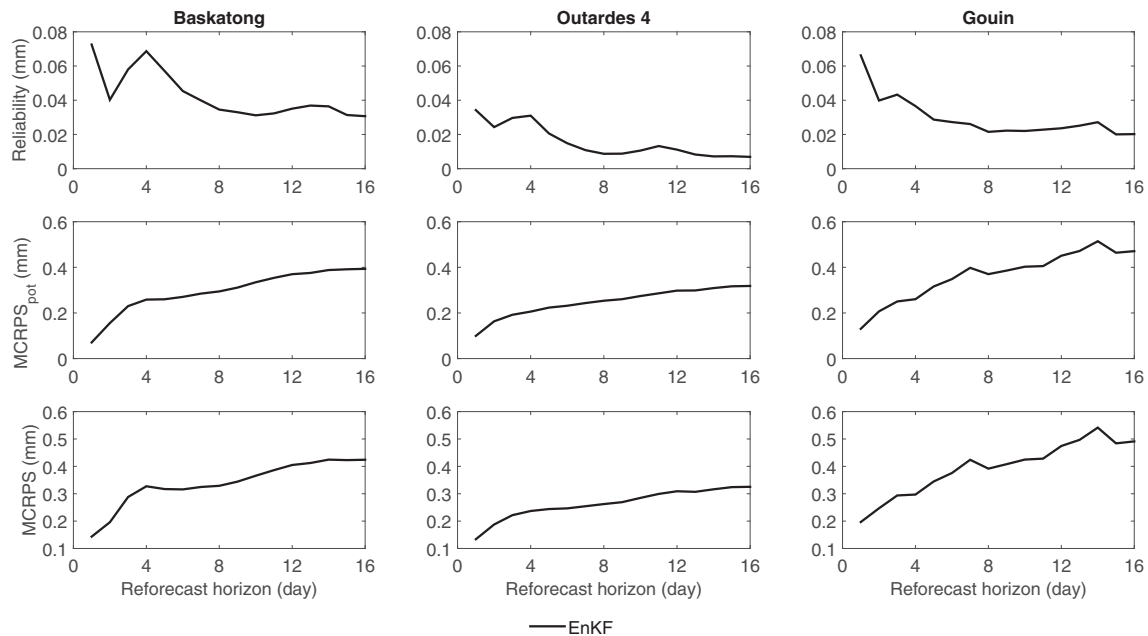


Fig. 8. Values of the MCRPS and of its two components for the hydrological EnKF scenario, as a function of the reforecast horizon.

first few reforecast horizons. As an example, in the Baskatong watershed, the reliability gain reaches almost 100% for day 1, but the $MCRPS_{pot}$ loss is larger than -100% , culminating in a total MCRPS loss of about -10% .

Identical conclusions to Fig. 9 were obtained using other reliability scores like the ratio δ presented in Fig. 10, which shows a substantial advantage to the use of the ensemble BMA hydrological processor (ratio δ values close to 1). Post-processing is indeed needed to achieve reliability since only two out of three sources of uncertainty are accounted for by the H-EPS: meteorological uncertainty and initial conditions uncertainty. In that regard, it does not matter if DBS is applied or not beforehand. On the other hand, DBS without ensemble BMA is insufficient to attain reliability, as it provides an under-dispersed meteorological ensemble for

which there is a higher probability of observations occurring outside of the ensemble. In most situations, the DBS + EnKF + ensemble BMA scenario offers the best results in terms of reliability. The reliability diagram for the case of DBS + EnKF + ensemble BMA scenario for Baskatong watershed is shown in Fig. 11 and confirms that the latter provides a reliable ensemble.

Fig. 12 shows that the DBS + EnKF scenario offers the best sharpness, with the lowest values of WPI in all reforecast horizons comparing to the other scenarios. The ensemble BMA hydrological processor cannot improve the sharpness compared to the EnKF scenario, but deteriorates the latter in most cases. The results obtained in Fig. 12 are also coherent with those of Fig. 10. Obtaining a smaller ratio δ , which is most desirable for achieving a reliable forecast as in the case of ensemble BMA (Fig. 10), has

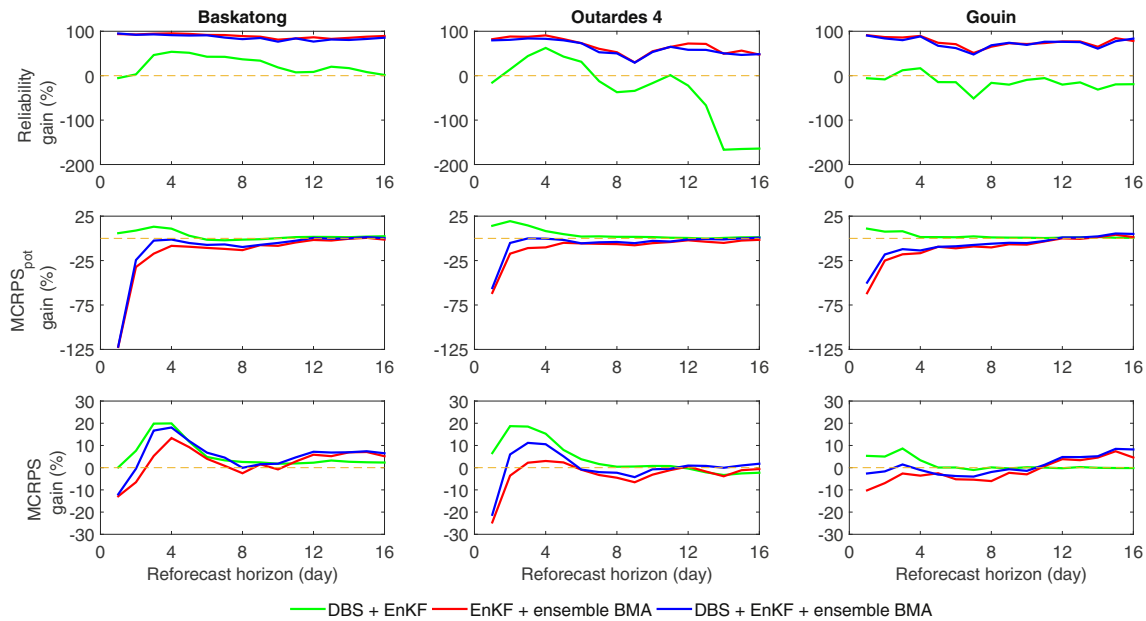


Fig. 9. MCRPS gains and MCRPS components gains (%) for the DBS + EnKF, EnKF + ensemble BMA and DBS + EnKF + ensemble BMA hydrological scenarios when compared to the EnKF scenario, as a function of the reforecast horizon. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

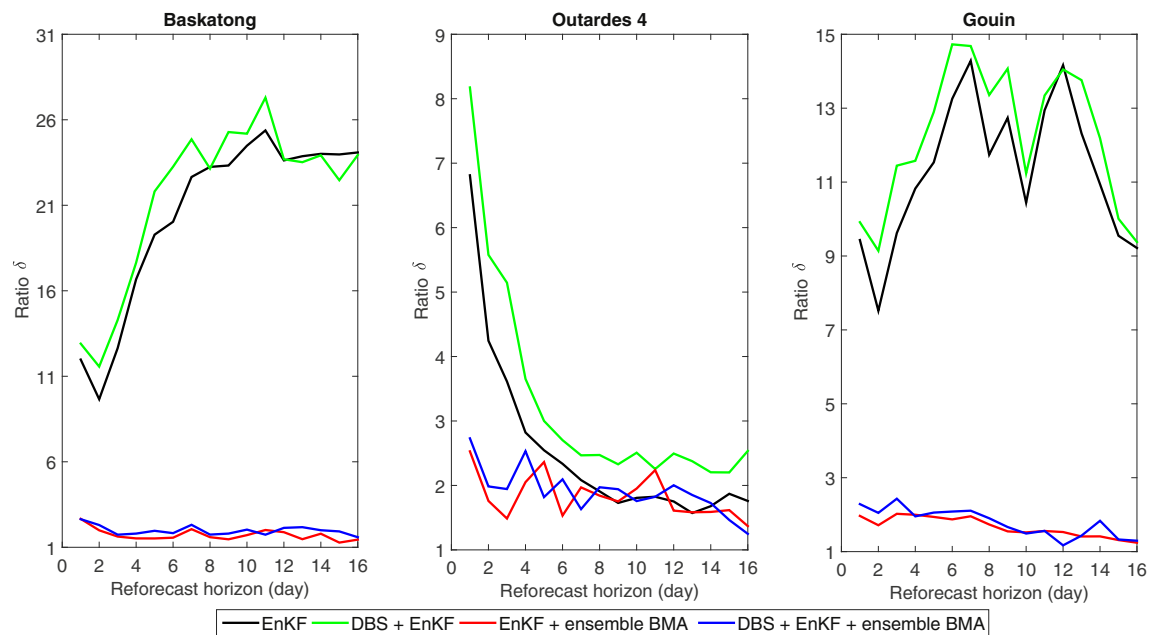


Fig. 10. Ratio δ values of all four hydrological scenarios, as a function of reforecast horizon.

caused the production of wide ensemble spread (high *WPI* values) which corresponds to an ensemble with lower sharpness or lower concentration of the predictive distribution.

Reduction of the bias of the meteorological ensemble forecast is clearly favorable to the hydrological forecast as the DBS + EnKF scenario outperforms all others (MCRPS), for most reforecast horizons. It thus seems that any bias error in the meteorological forecasts may only be efficiently corrected before being processed by a hydrological model. On the other hand, consideration of the ensemble BMA processor deteriorates slightly the MCRPS, probably because of the presence of extreme values in the randomly generated ensemble members (as in Wilks (2006a)). Overall, the DBS +

EnKF scenario constitutes the best option in terms of MCRPS, closely followed by the DBS + EnKF + ensemble BMA scenario. The same conclusions are observed in the all three watersheds.

Overall, the choice of scenario depends on the operational need, as the DBS + EnKF scenario offers the best resolution and sharpness while the DBS + EnKF + ensemble BMA scenario offers the best reliability. Some typical day 5 hydrographs for these two scenarios are presented in Fig. 13. It confirms that the ensemble spread (green shape) of the DBS + EnKF scenario is under-dispersed and that reliability is better achieved by the DBS + EnKF + ensemble BMA scenario. Note also that the mean of the ensemble (red line) is always quite close to the observations (black) in both cases.

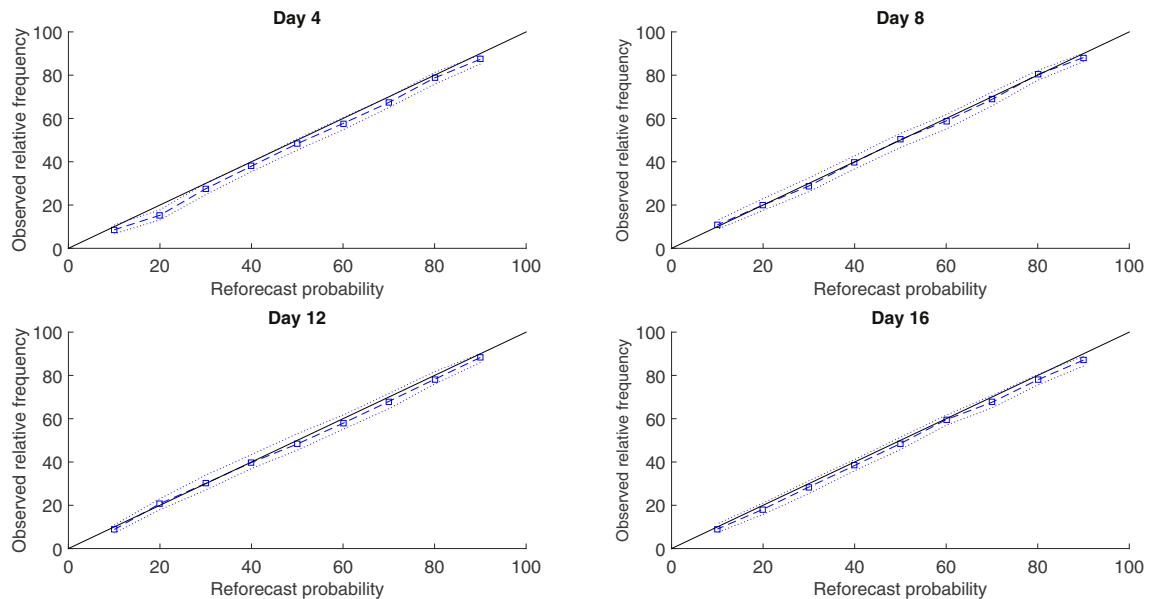


Fig. 11. Reliability diagrams for the DBS + EnKF + ensemble BMA scenario for the Baskatong watershed. The dashed lines depict the 90% confidence interval.

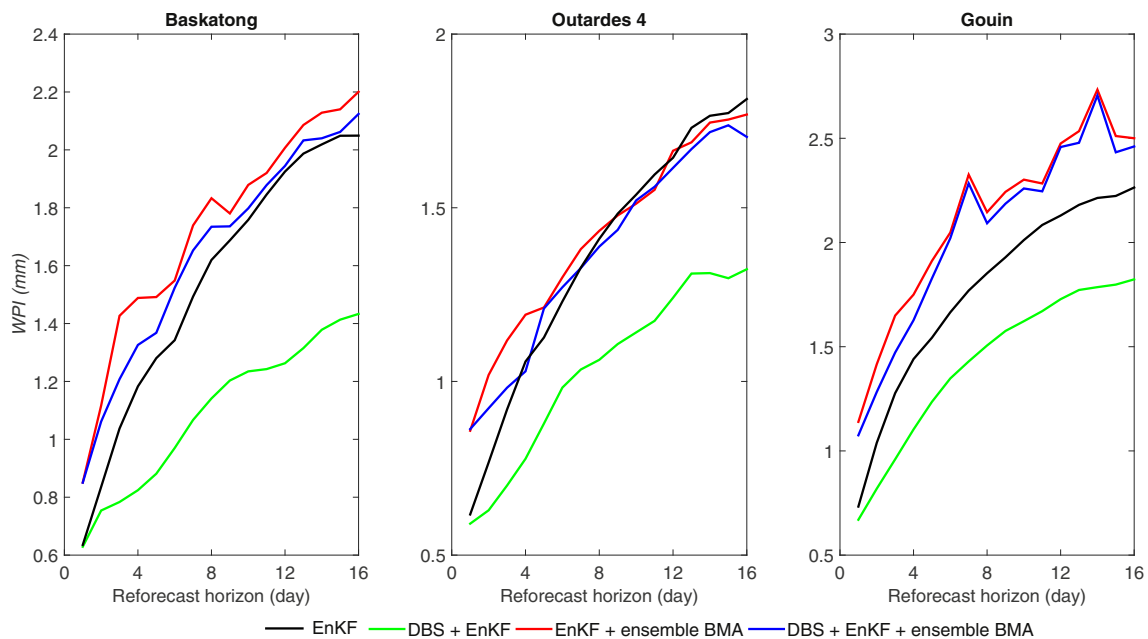


Fig. 12. WPI values of all four hydrological scenarios, as a function of reforecast horizon.

Resolution, sharpness and reliability are forecast qualities sought by all operational hydrologists. But one should also pursue extending the horizon of skillful forecasts: the number of days for which the H-EPS provides forecasts that are more informative than climatology. This issue may, for example, be explored through a skill score like the CRPSS. Fig. 14 shows that the DBS + EnKF + ensemble BMA scenario leads to skillful forecasts for all horizons and watersheds, since all CRPSS values are greater than zero. The large size of the tested watersheds certainly favored these excellent results: skills that possibly extend beyond 16 days. Similar results are obtained by the DBS + EnKF scenario (not shown). The selected scenarios can be implemented entirely or partly by forecasting agencies, depending on their objectives, needs, and operational constraints.

5. Conclusion

Meteorological processing is performed applying the DBS method, which focusses solely on reducing bias. Results show that it is beneficial to reduce the precipitation bias, for all seasons, but that it is detrimental to the spread of the predictive distributions, since DBS is not specifically designed to improve on this quality.

Hydrological processing is performed using the ensemble BMA method, which is specifically designed for multimodel applications such as this experiment, which relies on the combination of two meteorological reforecast products: the 4-member Canadian reforecasts from CCMEP and the 10-member American reforecasts from NOAA, leading to 14 members at each time step. Results show

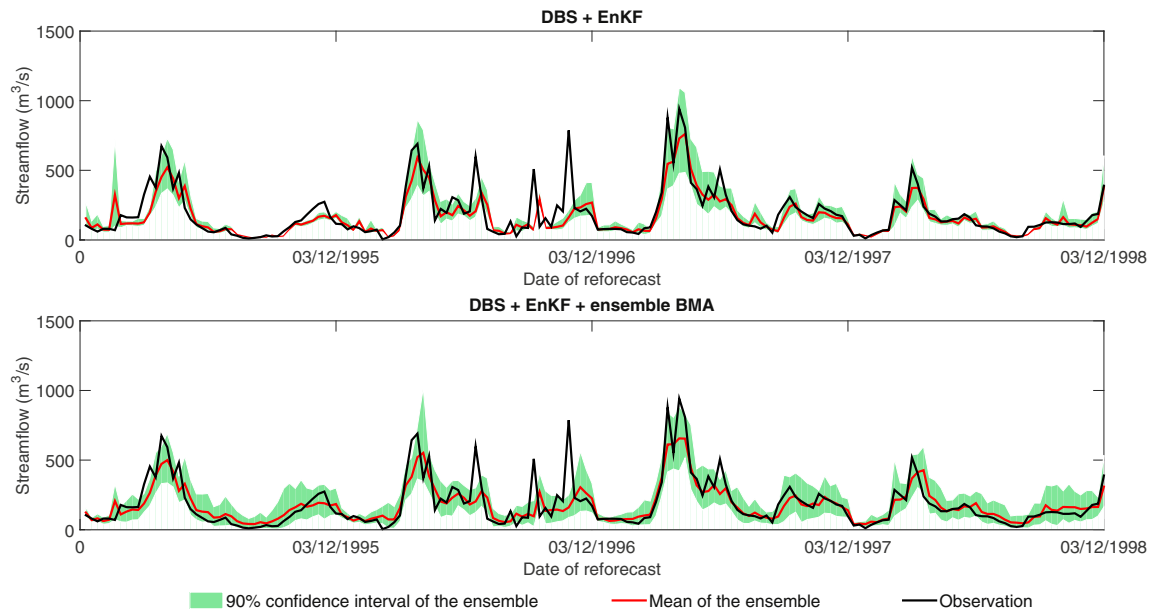


Fig. 13. Hydrographs of the DBS + EnKF and DBS + EnKF + ensemble BMA scenarios for 5-day lead time, Gouin watershed. The green shape depicts the 90% confidence interval of the ensemble, the red line illustrates its mean, and the black line the observation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

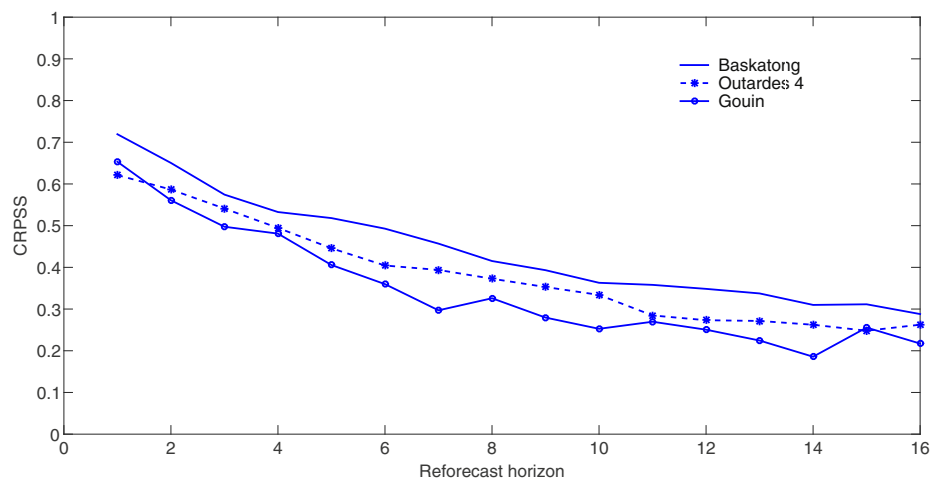


Fig. 14. Skill (CRPSS) of the DBS + EnKF + ensemble BMA scenario, as a function of reforecast horizon.

that the ensemble BMA succeeds in calibrating the spread of the hydrological predictive distributions.

As for the issue of combining or not the meteorological and hydrological processors, results show that all four tested H-EPS lead to small differences in terms of MCRPS resolution, with an advantage to the DBS + EnKF scenario. The ensemble BMA is thus unable to compensate for any bias left in the precipitation ensemble forecasts. Worst, some resolution is lost when the hydrological processor calibrates the predictive distribution. On the other hand, ensemble BMA succeeds in calibrating ensemble members that are otherwise under-dispersed. Therefore, if the reliability is preferred by users over resolution and sharpness, it is then the DBS + EnKF + ensemble BMA that should be retained, making use of both processors. On the other hand, if the resolution and sharpness is prioritized, the DBS + EnKF should be selected.

Many hydrologists, such as Schaake et al. (2007), have strongly advocated that it is important to process meteorological ensemble forecasts prior to their usage within an H-EPS. Many others, such as

Wilks (2006a), have shown the merit of a hydrological processor, especially for the common case when the H-EPS does not account for all sources of uncertainties and leads to under-disperse predictive distributions. Operational hydrologists are thus confronted with selecting one of the four options that are investigated here with an H-EPS that includes a probabilistic sequential data assimilation scheme (EnKF). Scheuerer and Hamill (2015) recently proposed a parametric statistical method that shares the quality of the DBS but not its flaws. It should be considered in subsequent studies.

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